F1 Race Intelligence — Final Project Report

Date: 1 November 2026 | Team ID: Not provided — enter the official assigned Team ID before submission | Project: F1 Race Intelligence: Formula 1 Performance Analytics using Tableau

1. Introduction — 1.1 Project Overview / 1.2 Purpose

F1 Race Intelligence is a data-visualization project intended to help users explore selected Formula 1 performance and reliability metrics. It uses a supplied Excel workbook, Tableau Public story and GitHub Pages website.

Purpose: turn structured race, qualifying, lap and championship records into visual comparisons for analysts, broadcast users and evaluators.

2. Ideation Phase — Problem Statement, Empathy Map, Brainstorming

Problem: manual comparison across several F1 data categories can be time-consuming and context can be lost. Primary personas: performance strategist, broadcast analyst and reliability analyst.

Chosen idea: a navigable Tableau story and project website. Alternatives considered: raw spreadsheet-only review, a live telemetry platform and a predictive model. The latter two were outside the verified project scope.

3. Requirement Analysis

Customer journey: define question → open story → compare visual → verify context → communicate finding.

Functional requirements: access the five workbook sheets, profile data, compare points/wins/podiums/poles, inspect qualifying/lap/reliability measures where present, publish the story, and link it from the website.

Non-functional requirements: clarity, read-only public sharing, traceability, responsive layout and documented limitations.

DFD: Excel workbook → data profiling → Tableau views/dashboard/story → Tableau Public → GitHub Pages website and report → users/evaluator.

Technology stack: Excel workbook, Tableau Public, HTML/CSS/JavaScript, GitHub repository and GitHub Pages. Snowflake is not verified as deployed.

4. Project Design

Problem–solution fit: analyst personas need quicker comparison and clear visual context. The solution hypothesis is a Tableau story supported by a public project site.

Architecture: workbook → quality profiling → Tableau authoring → Tableau Public Story1 → website navigation/README/report. Future Snowflake flow is proposed only.

Links: Tableau https://public.tableau.com/app/profile/sudhanshu.singh3051/viz/Book1_17936408624670/Story1?publish=yes | Website https://kmtp-png.github.io/Formula1-Race-Intelligence/#tableau | Repository https://github.com/en25it3010097-lgtm/formula-1-race-intelligence

5. Project Planning & Scheduling

Work packages: ideation and requirements; workbook profiling; Tableau views/story; website integration; documentation; testing and demonstration. Story points in the planning template are estimates, not a measured team velocity.

No daily task log was supplied, so actual burndown and actual sprint velocity are not claimed.

6. Functional and Performance Testing

Data profile: circuits 29 rows; championship_standings 311; lap_intelligence 3,814; qualifying_results 3,130; race_results 6,464; total 13,748 data rows excluding headers. Exact full-row duplicates: zero in each sheet.

race_results has 6,130 null total_race_time_s, 6,068 null dnf_reason and 396 nulls each in finish_position, positions_gained and gap_to_leader_s. These may be structurally meaningful and should not be blindly imputed.

Tableau tests to execute: public link access, labels/units, visible filters, tooltip/source reconciliation, mobile/desktop readability and measured load times. Snowflake execution tests are pending because no account/query evidence was supplied.

7. Results / Output Screenshots

The published Tableau Story1 and GitHub Pages website are the main public artifacts. Insert screenshots of the actual dashboard, each relevant story point, website Tableau link and any filter interactions before submission.

Do not include invented chart counts, fabricated query timings or unverified findings. Verify the exact chart inventory and any calculated fields from the saved Tableau workbook.

8. Advantages & Disadvantages

Advantages: public access, visual comparison, story-based presentation, traceable repository, and reusable data-quality notes.

Limitations: source provenance needs confirmation; some values appear curated/synthetic; missing fields require context; no live feed, automated refresh, verified Snowflake warehouse or predictive model is claimed.

9. Conclusion

The project combines a multi-sheet F1 dataset with Tableau storytelling and a public website. It provides a structured way to examine performance questions, subject to verification of chart details and dataset provenance.

10. Future Scope

Verify dataset provenance; add documented calculated fields and chart glossary; conduct usability testing; automate QA; connect to a governed warehouse if required; add live data only through a permitted and reliable API; explore prediction only after validating data and methodology.

11. Appendix

Source dataset: F1 Racing Project Dataset.xlsx (supplied workbook; five sheets). Tableau: https://public.tableau.com/app/profile/sudhanshu.singh3051/viz/Book1_17936408624670/Story1?publish=yes. Website: https://kmtp-png.github.io/Formula1-Race-Intelligence/#tableau. GitHub: https://github.com/en25it3010097-lgtm/formula-1-race-intelligence. Demo video: add the public Google Drive link after recording and sharing the narration video.

Author: Sudhanshu Singh. Team ID and any additional team members must be added using the official course/team details.

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